

MDS503-3 Java Programming

LAB2 Classes, Objects, Constructor Overloading

A retail company maintains an inventory system to track products stored in its warehouse. Each product may be registered at different stages with varying levels of information available.

Design a Java program to model and manage this inventory. Each product in the inventory has the following attributes:

- Product ID (integer)
 - Product Name (string)
 - Quantity in stock (integer)
 - Unit Price (double)
 - Reorder Level (integer)
1. Design a Java class to represent a product in the inventory.
The system should support creating product objects when partial information is available as well as when complete information is known.
 2. Store multiple product objects in an array of objects.
 3. Implement appropriate methods inside the class to support inventory analysis.
These methods should enable the system to:
 - Determine whether a product needs to be reordered
 - Evaluate the stock condition of a product
 - Compute total value of the product.
 4. Use suitable control structures and loops to:
 - Display complete details of all products
 - Identify products that require attention based on stock levels
 - Determine which product contributes the highest value to the inventory
 - Compute summary statistics that would be useful for warehouse management